

DESIGN (E) 314
TECHNICAL REPORT

Smart Prepaid Energy Controller

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1 Introduction and system description (3)

This report covers the design of a smart energy meter, It monitors and controls the electrical load to a tenants housing. Monitoring and control will be authorized by the landlord through the meter.

2 Hardware design 15

The following descriptions are visually shown in Figure 1. A set of different components are used to achieve the above mentioned outcome. The energy meter will use an STMN32GOB1RE board with a NUCLEO-64 form factor. The STM board would read energy values, log them to an SD card, confirm user authentication with and RFID, manage menus with UART and an OLED, have warning and debug LEDs and use a keypad for user friendly input. To power all the components on the student development board, a single +9V input is supplied to the student board. The load control circuit and STM board would require +5V to operate, with the STM power selected to E5V. The rest of the components would require a +3.3V power supply. Thus, two power regulators are used, a +5V regulator and a +3.3V regulator. The keypad and debug LEDs are the only components powered through the STM board itself. The reading of the loads voltage and current values will be through the STM boards ADC. Since the ADC can only take a range from 0V to 3.3V, the signals would have to be conditioned. This is done using a signal conditioning op-amp circuit. It converts the pure sinusoidal signal to have some positive gain and offset so that the input signals to the system are conditioned to be within the prescribed ADC voltage limits.

The load to the tenant housing will be simulated with a Load Relay circuit, the Relay will be enabled by the stm board and a physical push button(S3) . The current used to turn on the coil would be too high if routed through the STM board, thus a MOSFET in the relay control circuit would be used to drive the relay open or closed.

The SD CARD and SD CARD reader will be used to log energy metering data to maintain the energy usage history. Even when there is a power outage, the data would remain on the SD card ensuring a reliable record of values are stored. To only allow authorised operation of the energy metering device, an RFID reader and card or tag will be used. Both the RFID reader and SD CARD reader will communicate with the STM via SPI. Only the OLED(indicated in purple) will be using I2C as a communication protocol.

For UART communication, the Test Interface Connection (TIC) will be connected to the RX and TX pins of the board. Alternatively for development and testing the USB connection can be used on the STM board. All other TIC connections are indicated clearly in appendix A.

The keypad will be used to navigate menus viewed on the Oled and enter the number of units to add. The keypad rows and columns would be scanned for button presses using the GPIO of the STM pins only. The keypad is not connected to anything else such as the power supply. A hardware block diagram implementing the function of the energy meter is shown in Figure 1, a full wiring schematic is shown in appendix A. A full list of pins used and their configurations are shown in Appendix B.

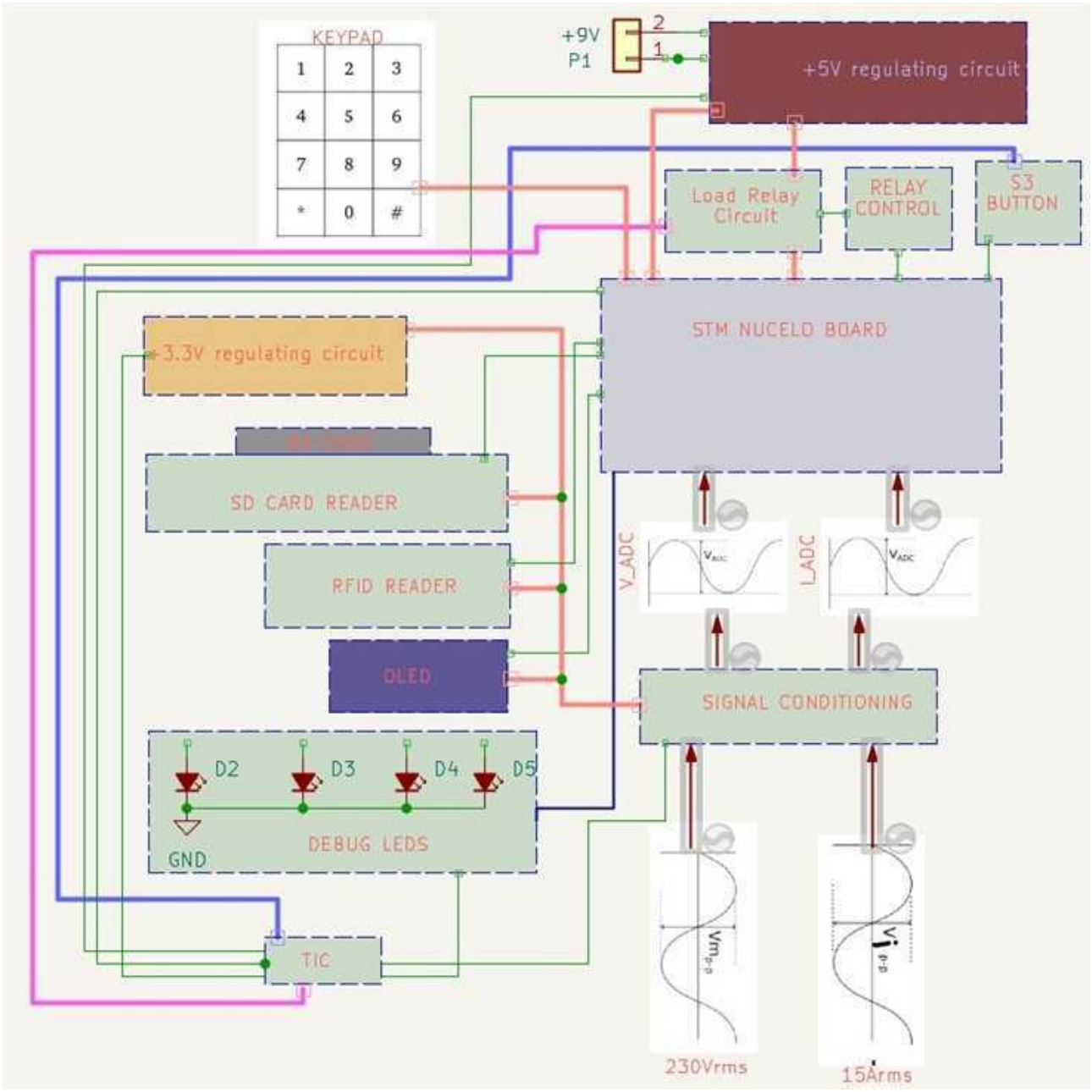


Figure 1: Hardware block diagram

2.1 Power supply

The two power supply regulator circuits were designed to provide reliable +5V and +3.3V voltages. The design for the +5V regulator supply was provided by the E-Design lecturers. For a +3.3V regulator, a L78L33 in a T0-92 package. [1] provides a recommended circuit. From the recommended circuit, the input capacitance is unnecessary since the output of the +5V regulator already uses 2 parallel capacitors. Their equivalent capacitance is already 10.1uF. The extra 0.33uF would only change the capacitance to 10.43, thus it can be ignored. The final implemented 3.3V regulator can be seen in appendix A. From [1], it can be seen that the L78L33 can be operated with a junction temperature between 0 and 125 degrees Celsius. At 25 degrees the typical output voltage is +3.3V. For an input voltage of 5.3 to 20V, and an output current between 1 and 40mA, the output voltage ranges between 2.97 and 3.63 Volts. The Range should be expected to be some 0.1V less since we are supplying 5V to the LM7L833. The implication of the change in output voltages here is asessed by looking the components that make use of the -3.3V regulator.

The output voltage range of the L78L33 regulator remains suitable for all connected components. According to the RFID reader's datasheet [2], it accepts a supply voltage between 2.5V and 3.3V. The OLED module, as shown in [3], supports a wide input range from 2.5V to 5.5V. Similarly, the SD card reader, based on the datasheet from STEMlearn [4], operates between 2.6V and 3.6V. The L78L33's output—typically 3.3V and ranging between 2.97V and 3.63V under a 5-20V input—is well within the acceptable range for all three peripherals. Since the input voltage to the regulator is well below its maximum rating of 20V, any minor deviation in output (e.g., up to 0.3V) is considered negligible.

As long as the 3.3V output remains stable, it will not affect the behavior of any connected analog components, including the op-amp stages. This is because the power supply variations are already mitigated by the use of smoothing capacitors on the outputs of the 5 and 3,3V regulators, which help filter out noise and ensure voltage stability. With these conditions met, the overall system can be expected to operate reliably and within specifications.

2.2 LEDs Debug

The recommended operating current for the red LEDs used in the lab is 20mA, according to [5]. The typical voltage supplied to the LEDs is 3.3V from the GPIO pins. Using the minimum forward voltage, the required series resistance to limit current can be calculated as:

$$R = \frac{V_{\text{GPIO}} - V_{\text{LED,min}}}{20\,\text{mA}} \tag{1}$$

$$= \frac{3.3\,\text{V} - 1.9\,\text{V}}{20\,\text{mA}} \tag{2}$$

$$= 70\,\Omega \tag{3}$$

The STM is running multiple peripherals at once. Assuming all LEDs are on required 80mA to run from the STM board to LEDS alone. Limiting the debug leds to 5mA per LED brings their resistor values to 280 Ohms. The closest available resistors are used an implemented in the final design and shown in appendix appendix A. The final designed value for the LEDS remained at:

$$R = 390\,\Omega \tag{4}$$

2.3 V/I sensing circuits using gain and offset op-amp circuit

The E Design 314 2025 project requires measurements of a 230 Vrms voltage and a 15 Arms current. For safety, both signal values will be simulated using 2V peak-to-peak, centered around 0V. Because the STM board's ADC requires a positive input voltage (0 to 3.3V) [6], a gain and offset amplifier is necessary to condition the simulated signals. Finally the accuracy needs to be within ±5% the actual value.

To maximize measurement accuracy, the amplifier will aim to utilise the full ADC input range (0-3.3V). Dual MCP602 op amps were provided and will provide the necessary gain and offset for both

the simulated voltage and current signals. Resistor values for the op-amp circuits were calculated using the "Designing Gain and Offset in Thirty Seconds" [7] method. From [7], the parameters required for the op-amp are: (Output full scale voltage (Voutfs); Output zero scale voltage (Voutzs); Input full scale voltage (Vinfs); Input zero scale voltage (Vinzs); and Voltage level of stable reference (Vref) . [8] shows that for linear output voltage swing, The output voltage should be between Vss+100mV and Vdd-100mV. This corresponds to 0.1-3.2V. Since a ±5% tolerance needs to be accounted for, the design should allow for a 110% of the output voltage in the 0.1-3.2V range. This is demonstrated in Figure 2 For safety and expectations of discrete resistor values, a range of 0.16-3.14 is chosen instead and displayed in Table 1. The input voltages are also compensated for the ±5% accuracy requirement. These values are used to calculate m and b from equations of [7] shown below. The op amp from [7], shown in Figure 3 is realized with the MCP602 op-amp, shown in Figure 4 and it's implementation is implemented in the full schematic shown in appendix A

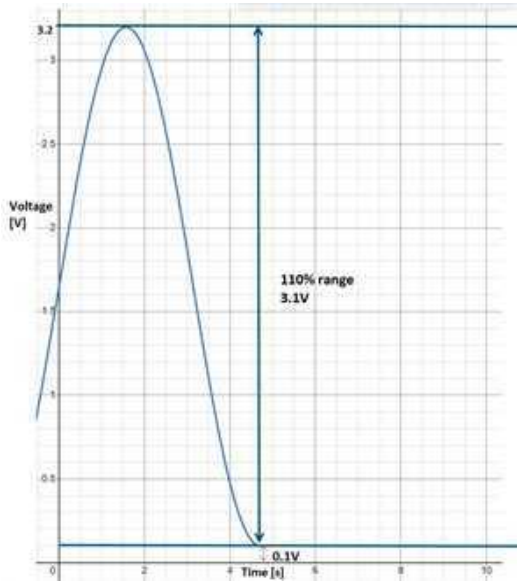


Figure 2: Safe operating range

Variable	Value	Justification
Voutfs	3.14V	Compensation for Voutfs ×105%
Voutzs	0.16V	Compensation for Voutfs ×5%
Vinfs	1.05V	Compensation for ±5%
Vinzs	-1.05V	Compensation for ±5%

Table 1: Signal Conditioning Parameters with Technical Justifications

$$m = \frac{Voutfs - Voutzs}{Vinfs - Vinzs} \tag{5}$$

$$b = Voutzs - m \cdot Vinzs \tag{6}$$

$$R_2 = \frac{V_{\text{ref}} \times R_1 \times m}{b} \tag{7}$$

$$R_g = \frac{R2 \times R_f}{m \times (R1 + R2) - R2} \tag{8}$$

The values of m and b then equals 1.418 and 1.65V respectively . This is a positive m and and positive b. [7] Gives the corresponding circuit to implement the op-amp for this configuration in Figure 3. The corresponding resistor values are then chosen and calculated as follows: R1 is chosen to be 12kOhm; thus R2 becomes 34.05 kOhm; Rf is chosen to be 9kOhms and Rg then becomes 9.79kOhms. The closest resistor values in the lab were then chosen.The closest possible resistor value for Rf was ahcieved only through two resistors in series. RF is split into RF#1 and RF#2 where the # stands for A or B after each resistor. A indicates the Voltage(A) op-amp or the Current(B) op-amp. The closest E24 standard available resistors in the engineering labs were chosen and their final real

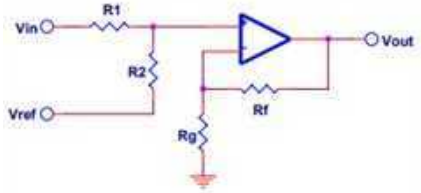


Figure 3: Schematic for positive m and b.

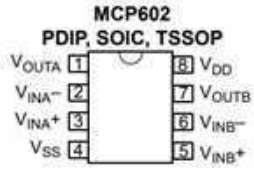


Figure 4: Data Sheet Schematic

values are shown in Table 2. The effects of change in Resistor values are investigated in Measurement and results later. The exact same aproach for implementing the Voltage signal coniditioning was used for the current signal conditioning.

Resistor	**Calculated**[Ω]	**Available**[Ω]	**Real** [Ω]
RF1A	split RFA	8.2k	8.1k
RF2A	split RFA	820	803
RGA	9.78k	10k	8.69k
R2A	34.05k	33k	32.2k
R1A	12k	12k	11.76k
RF1B	split RFB	8.2k	7.58k
RF2B	split RFB	820	806
RGB	9.78k	10k	8.70k
R2B	34.05k	33k	32.2k
R1B	12k	12k	11.73k

Table 2: Op-amp Resistors

2.4 Load Control Circuit

For the simulated load, a J1071AS5VDC36 relay was used to represent a controllable power path. When the relay switches off, a large negative charge can be generated over the coil. To protect the realy coil, a 1N4007 diode is connected in parallel with the coil terminals. The +ve diode terminal is connected to the -ve relay terminal to allow safe dissipation of flash negative voltages. It can be seen from [9], for an input of 5V to the coil, the relay would switch on at 3.75V and off at 0.3V. The current required to drive the coil is 72.5mA. Both parameters cannot safely be provided by the STM board. Thus, a 2N7000 N-Channel MOSFET was implemented as a low-side driver. The datasheet [10] shows that the minimum on drain current when is 75mA. The MOSFET's gate is controlled by a GPIO pin which in turn controls the Relay.; when the GPIO is set high, the MOSFET conducts, allowing current to energize the relay coil and complete the simulated load circuit. To Protect the STM and accidential energising of the relay, resistors R4 and R3 need to be used. For the R4, when the GPIO pin transitions high, it charges the MOSFET's gate capacitance. Without R4, this could be a very high instantaneous current, potentially damaging the GPIO pin or causing ringing. R4 limits this inrush current.

$$I_{\text{peak_gate}} \approx \frac{V_{\text{GPIO_HIGH}}}{R_4} \tag{9}$$

Substituting the given values:

$$I_{\text{peak_gate}} \approx \frac{3.3\,\text{V}}{1\,\text{k}\Omega} = 3.3\,\text{mA} \tag{10}$$

This peak current is well within typical GPIO pin capabilities. However the peak gate voltage can is 0 at steady state and thus not a concern for the GPIO pin. When the STM board is powering up R3 pulls the MOSFET's gate voltage down to ground (0V). This ensures that VGS is 0V, keeping the MOSFET OFF and the relay de-energized, preventing accidental activation.

The same GPIO pin used for LED D2 was also connected to the gate of the MOSFET. This ensures that LED D2 lights up whenever the relay is activated, providing a visual indication that the simulated load is being powered. With the forward voltage as 1.9V, and choosing a reasonable 1kOhm resistance, the current in through the LED bcecomes:

$$I = \frac{V_{GPIO} - V_F}{R} = \frac{5\,\text{V} - 1.9\,\text{V}}{1000\,\Omega} = \frac{3.1\,\text{V}}{1000\,\Omega} = 3.1\,\text{mA}$$

This is below the recommended LED current and thus is safe for operation. The load circuit and control schematic is shown in Figure Figure 5 and it's full implementation shown in appendix appendix A

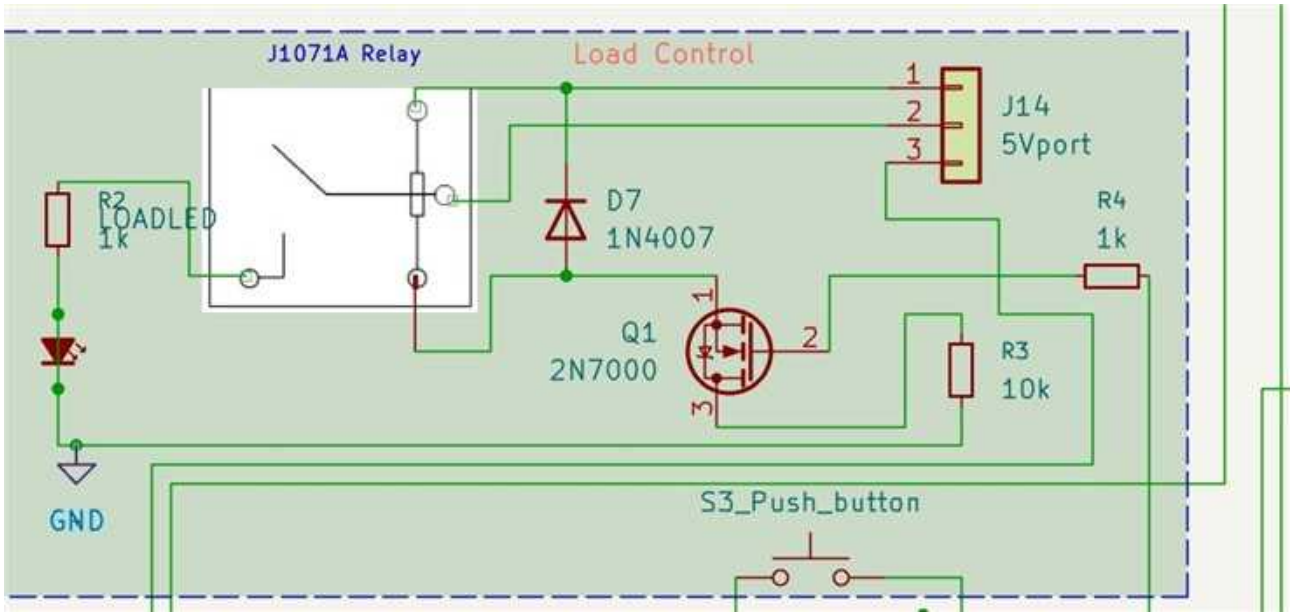


Figure 5: Load Circuit and control schematic

2.5 Keypad and button

The keypad will be used to navigate menus and input units to the system. The keypad implemented had 4 (rows) x 3 (columns). A multimeter was used to verify each row and column connection. Other keypads may have different configurations and need to be similarly verified. The typical layout is indicated in-Figure 6.

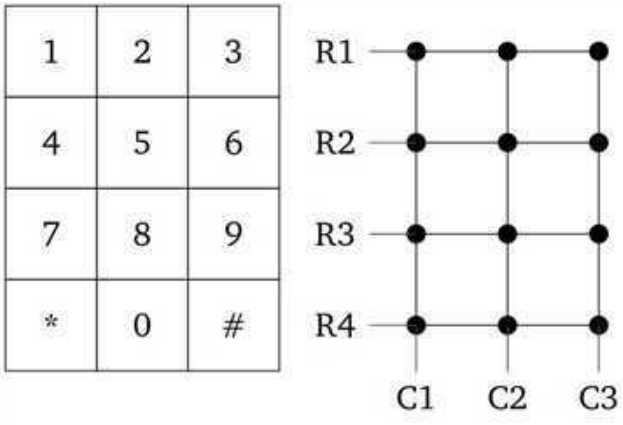


Figure 6: KeyPad Layout

The rows have been configured as GPIO high 3.3V outputs and the columns as GPIO External interrupts on the micro-controller. The S3 button will be used for emergency load control. It will toggle the load on/off with each button press. This will allow a user to test the load control and to override the control if needed. [3].The button emergency was configured to have 3.3V from a jumper connected to one end of the button and the other end connected to pin PD9 and configured as an interrupt. The push button will be used to override the load control. This can be useful during emergency situations.

The S3 button node connect to PD9 is also connected to the TIC. Their detection and debouncing are covered in the software section later. The wiring is shown in appendix A. The pinout configurations are shown in table 3

Table 3: Keypad GPIO Pin Mapping

Rows (Output)	**STM32 Pin**
Row 1	PB2
Row 2	PA11
Row 3	PC1
Row 4	PC0
Columns/S3 (Input)	**STM32 Pin**
Column 1	PB6
Column 2	PB10
Column 3	PC0
S3	PD9

2.6 oled

The OLED that was used had the following specifications:

- Interface: I2C
- Resolution: 128 × 32
- Diagonal Screen Size: 0.91”
- Display area: 7mm × 25mm / Board Size: 12.0mm × 39mm
- Thickness: 4mm
- Pixel Size: 0.159mm × 0.159mm
- Technology: OLED Blue
- Graphic Driver IC: SSD1306 (This determines the software approach)
- Wide input voltage range: 2.5V–5.5V

[3] The typical onnections are shown in table 4

Table 4: OLED I2C Connections

OLED Pin	**STM32 Pin**	**Connection Description**
VCC	3.3V Jumper	Power Supply
GND	GND	Ground
SDA	PB9	I2C Data Line
SCL	PB8	I2C Clock Line

The wiring is shown in appendix A.

2.7 SD Card

The connections made are shown in table 5

Table 5: SD CARD CONNECTIONS

SD CARD PIN	Conection
VCC	3.3V Jumper
GND	GND
MISO	PA8
MOSI	PA7
CLK	PA5
CS	PB0

2.8 RFID reader

When designing the RFID placement. Testing different position that reduces noise on the power supplies ensures that the ADC readings do not change or distort due to the RFID and produce accurate readings. Suitable placing was tested and shown later in results and testing. The wiring is shown in appendix A. The connections made are shown in table 6

Table 6: RFID CONNECTIONS

SD CARD PIN	Conection
VCC	3.3V Jumper
GND	GND
SDA	PD2
SCK	PC10
MOSI	PC12
MISO	PC11
IRQ	Not Connected
RST	PD0

The wiring is shown in appendix A.

3 Software design and implementation

3.1 Software logic and interaction, Musa insert flow diagram (2)

The Software implementation is explained below. The implementation is in accordance with the Flow and FSM diagram in Figure 7. For clarity the OLED menus are displayed separately in Figure 8. 1. Signal Acquisition and Processing Conditioned analog signals (e.g., voltage or current) are continuously sampled by the ADC (Analog-to-Digital Converter).

Once a sample is ready, an ADC complete callback sets a flag (adcready = 1) indicating that new data is available.

In the main loop, when adcready is set, voltage conversion and power calculations are performed.

If the apparent power exceeds a defined maximum threshold, a warning LED is flashed.

2. User Interaction and Display A button press is detected and triggers updates to the OLED display and navigates the menu system.

The system monitors the remaining energy units, and if units fall below a threshold, a warning LED is also triggered.

Periodically, based on time and a logging flag, energy data is logged to the SD card.

3. RFID-Based Access and Control The system listens for UART interrupts, which indicate the reception of commands (e.g., from an RFID scanner).

When a UART command is received:

It may transmit current energy info, toggle the load ON/OFF, or add energy units.

The system confirms RFID authentication, and if successful, performs the requested operation (e.g., ADD XX units).

4. Data Logging and SD Card Interface If the LOG FLAG is set and a time condition is met, the program logs data to the SD card.

The user can also issue a command via UART to clear the SD card or read back the log data for transmission over UART.

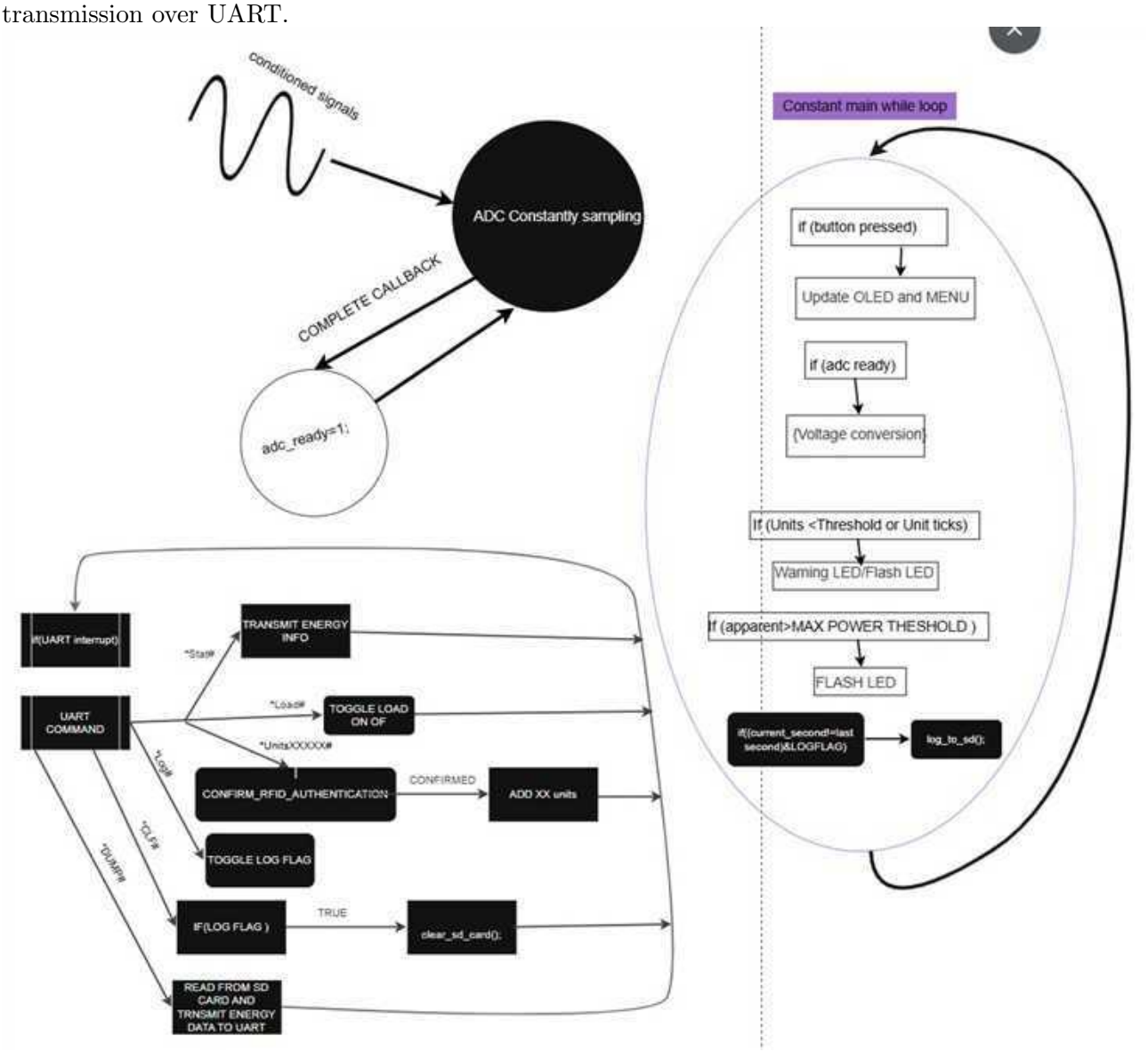


Figure 7: Software FSM and flow diagram.

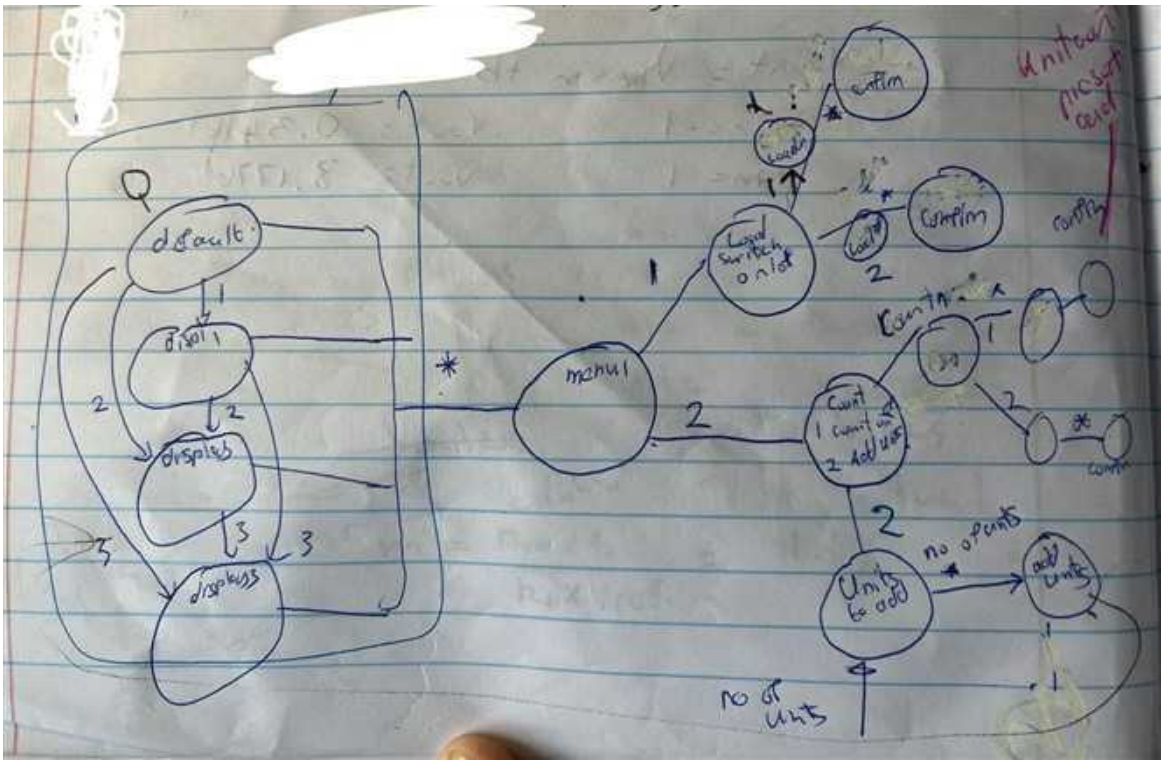


Figure 8: OLED FSM and flow diagram.

3.2 ADC, Setup and sampling data management (1)

A 12-bit Analog-to-Digital Converter (ADC) was selected, providing a resolution of:

$$\text{Resolution} = \frac{3.3\text{ V}}{2^{12}-1} = \frac{3.3\text{ V}}{4095} \approx 805.86\ \mu\text{V/LSB}$$

This is more than sufficicent for the accuracy requirement To convert raw ADC readings to calibrated voltage values, the following formula is used:

$$V_{\text{actual}} = \text{ADC}_{\text{value}} \times \frac{V_{\text{out,FS}} - V_{\text{out,ZS}}}{\text{ADC}_{\text{FS}} - \text{ADC}_{\text{ZS}}}$$

The RMS Voltage is then calculated as:

$$(V_{\text{maxactual}} - V_{\text{minactual}}) \times \frac{V_{\text{out,FS}} - V_{\text{out,ZS}}}{\text{ADC}_{\text{FS}} - \text{ADC}_{\text{ZS}}}$$

To achieve the required 5% accuracy for the 50 Hz signals, a careful determination of the sampling rate is necessary. A 5% tolerance over the full 360° cycle of the 50 Hz sinusoidal input translates to an angular resolution of 18°. However, the greatest rate of change in the sinusoidal signal's magnitude occurs near the zero-crossing point. A 5% amplitude variation around this point corresponds to an angular range of ±1.433°, resulting in a critical angular window of 2.866° which is less than the 18°s. To capture this variation accurately, the sampling period must be no greater than 159.222 µs (2.866°/360° × 1/50s). Considering the need to sample both channels, the effective sampling period per channel is halved to 79.62 µs, dictating a minimum sampling rate of approximately 12.56 kHz. Sufficient margin is then designed in the ADC discussion below to ensure the ADC and associated timer system can reliably capture the required data while maintaining the desired accuracy. The chosen timer frequency, which is significantly higher than the required 15 kHz sampling rate, ensures ample processing capacity.

The analog-to-digital conversion (ADC) system is designed to sample both the voltage and current channels concurrently. The ADC employed has a 12-bit resolution and requires 15 ADC clock cycles per conversion (12 cycles for settling time + 3 cycles for conversion)[6]. Since current and two channels

are sampled since 30 cycles are needed. The closest allowable clock cycles of 39.5 is then chosen. The ADC clock is run by the PCLK,is synchronous and derived from the system clock (SYSCLK). The SYSCLK at 64 MHz and the APB prescaler set to /1 and the ADC set to to Synchronous clock mode divided by 2. The SYSCLK frequency is chosen to be 64 MHz. Therefore, the ADC clock operates at 32 MHz (64 MHz / 2). This results in a maximum conversion rate of 810 kHz (32 MHz / 39.5 cycles). This is 64.5 (810khz/12.56kHz) times faster than what is needed to accurately sample. Thus the clock frequency chosen is suffficiently fast for an accuracy of 5% at 50Hz. The flow diagram of the Hardware and Software implementation is shown in Figure 9

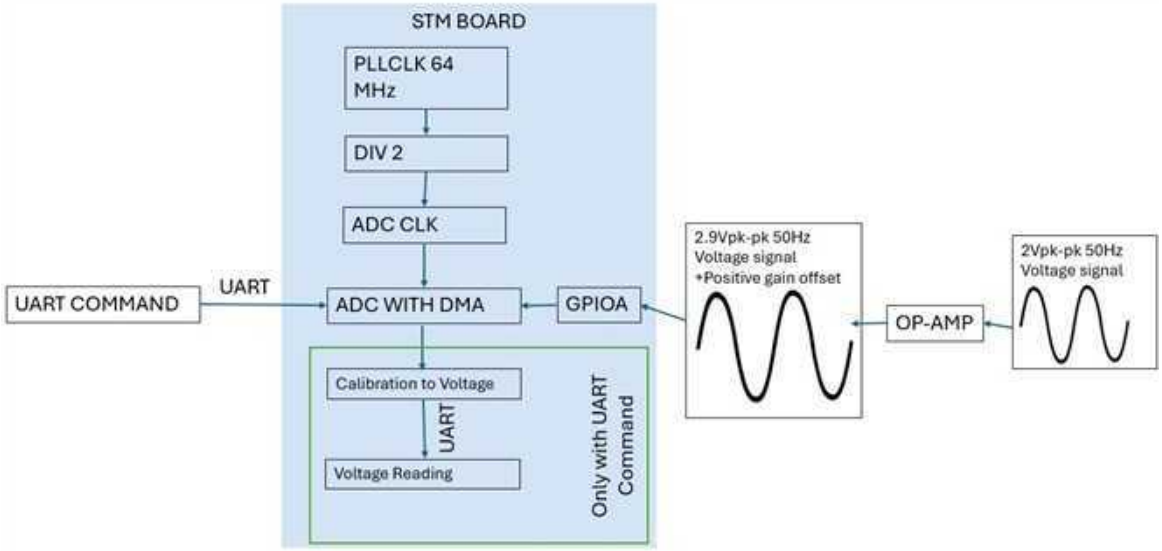


Figure 9: Software flow diagram.

3.3 Analogue measurements, voltage, current, and phase (2)

3.4 Power calculations, P, S, Q, and PF (1)

3.5 UART communications (protocol, data conversion) (2)

3.6 Keypad driver setup and button bounce handling (1)

3.7 Keypad driver setup and button bounce handling (1)

3.8 OLED low-level interface (1)

3.9 FSM control and logic (Inputs: Keypad, UART. Outputs: LEDs, Load, OLED) (4)

3.10 Energy consumption calculations + RTC (1)

3.11 SD Card Interface and Logging (2)

3.12 RFID Interface and authentication (1)

4 Measurements and Results 14

• The intention of this section is to demonstrate that the built hardware will perform the functions as defined by your hardware design requirements. • For each element, provide the method that was used to verify the functionality or perform measurements, and report on the result (measurement results, and/or successful verification).

4.1 Power supply

4.2 V and I OP-amp circuit

Extensive testing showed that the Voutfs of the amplifier remained between 3.18-3.24V for a 2Vpk-pk signal. A DC test showed that the Voutfs remained between 3.12-3.16V.Voutzs stayed between 0.319-0.36V for ac and dc testing respectively. This is demonstrated in Figures 10 and **??**. Substituting the resistor values for the voltage op-amp back into equations 2,3 and 4 we find that the new values for m and b are 1.46 and 1.76 respectively. Voutfs and Voutzs is then determined to be 3.178 and 0.34V respectively. The differences between the calculated new values and the practical values are on the order of 10^{-3} Volts. For the practical voltage range of (choosing closest ac values) 3.18-0.319$\times$0.05 = 0.14V which is 100$\times$ larger then the 10^{-3} Volts difference. Therefore the actual Voutfs and Vouzs values are sufficient for the $\pm$5 accuracy requirement.

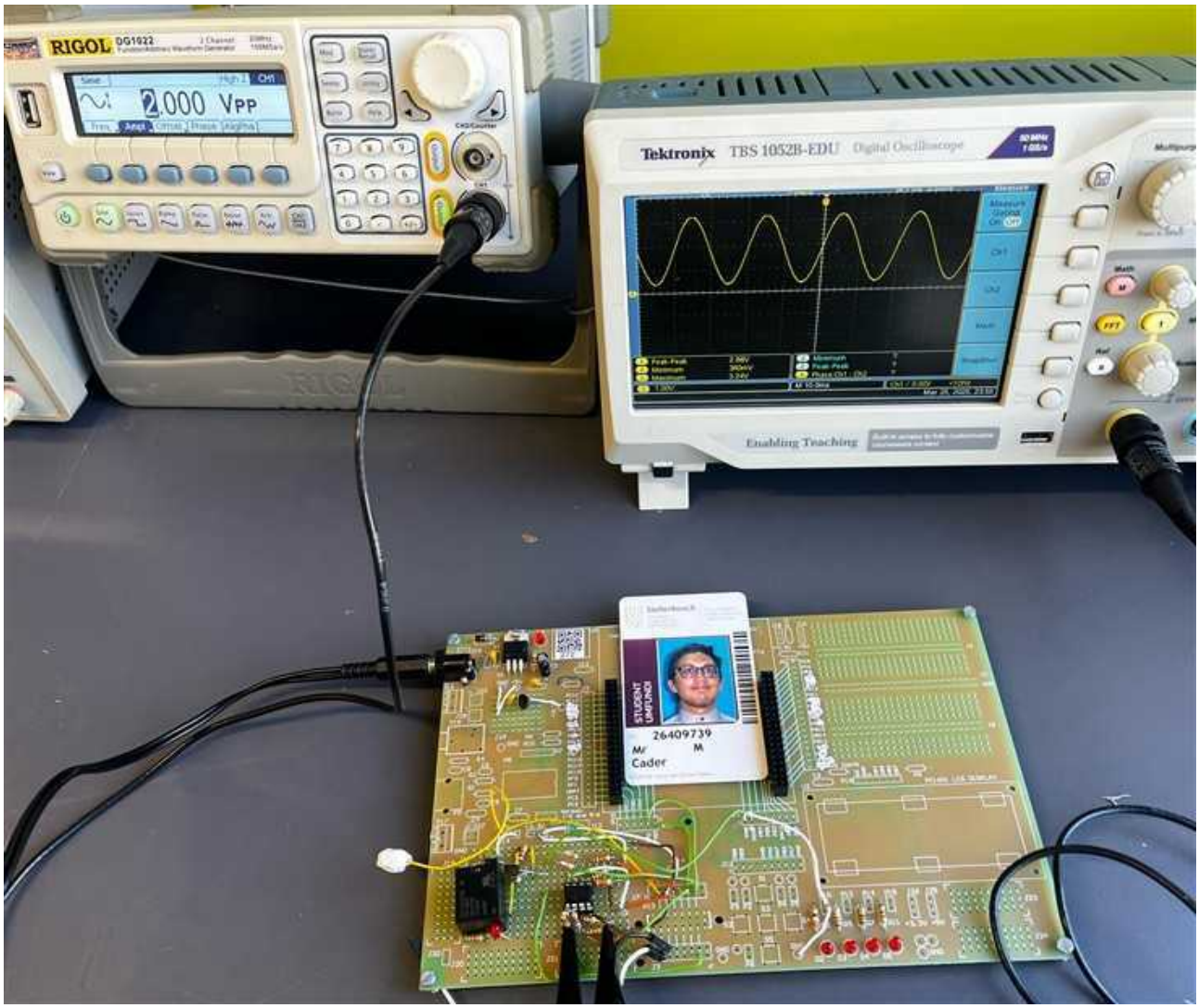


Figure 10: Measured op-amp

4.3 Load circuit

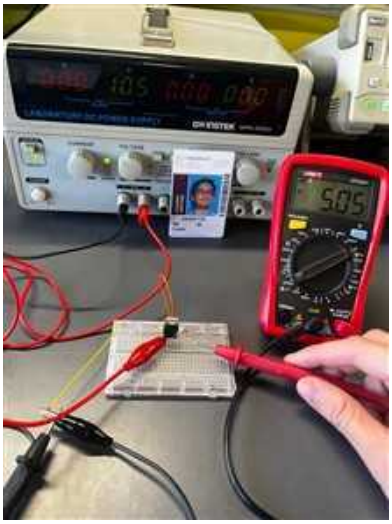


Figure 11: 5V regulator with 10.2V input showing 5.02V output (Test performed with Aaadam Noordien)

4.4 Keypad and button



Figure 12: Keypad Rising edge



Figure 13: Keypad Falling edge

4.5 UART communications

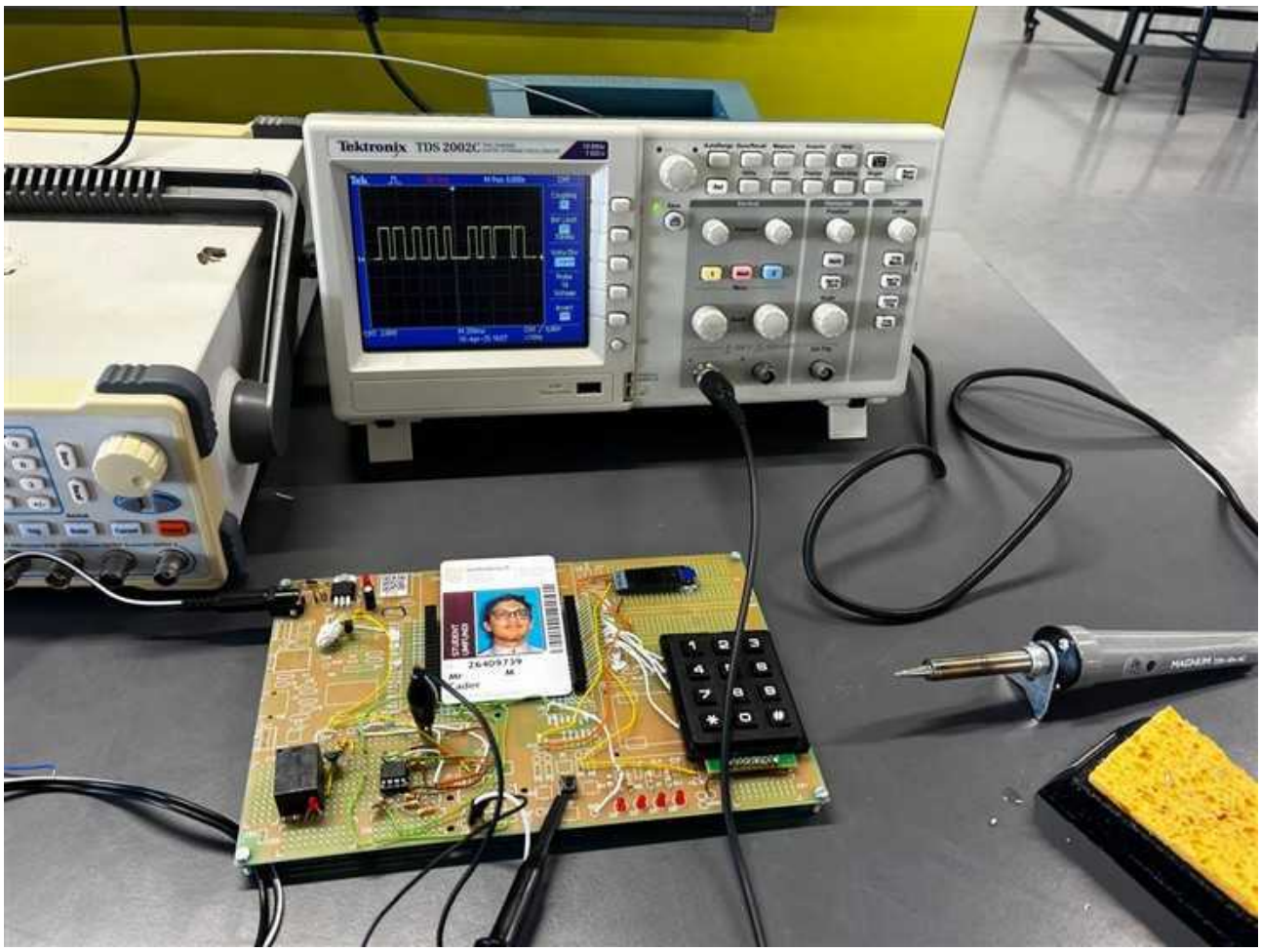


Figure 14: S3 button signal

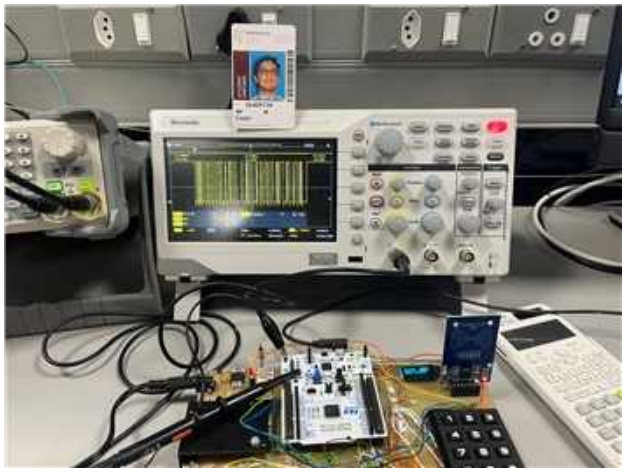


Figure 15: Uart Tx Student Number

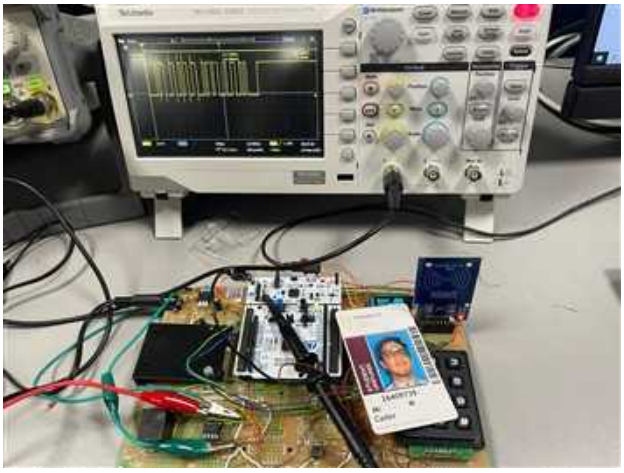


Figure 16: UART Rx CLF command

4.6 V, I, Phase, P, S, Q (incl. calibration)

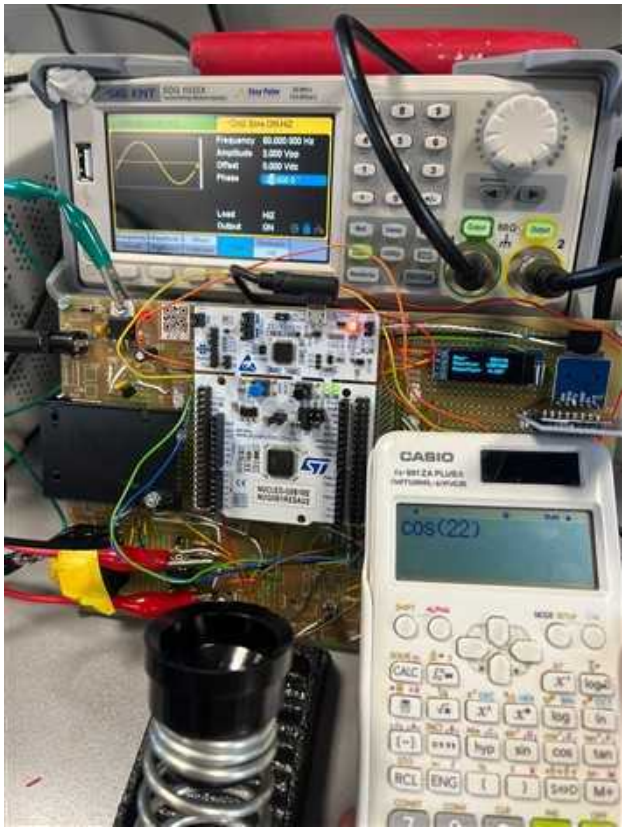


Figure 17: Phase calibration test

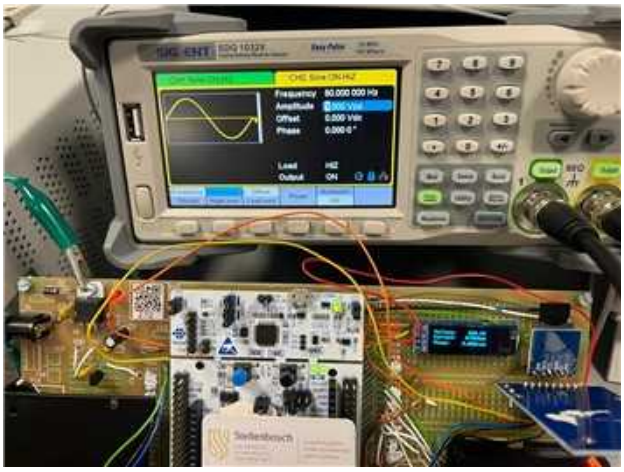


Figure 18: OLED output Voltage at 1Vpp test

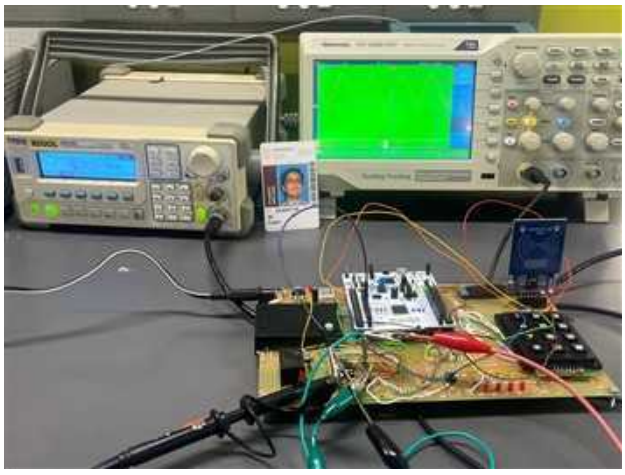


Figure 19: Voltage signal with RFID

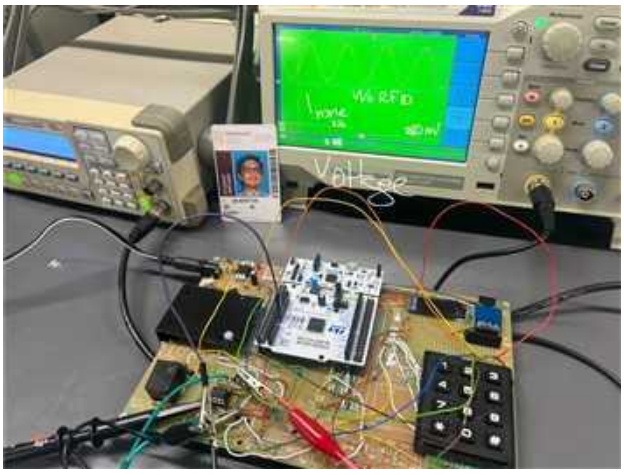


Figure 20: Voltage signal without RFID

4.7 OLED (functional)

4.8 SD Card

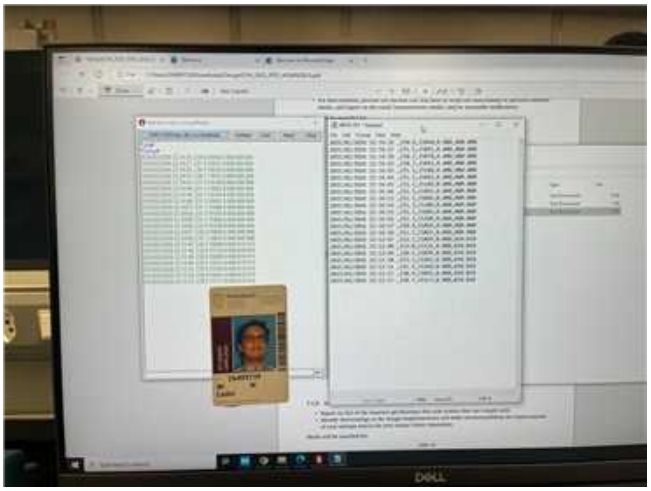


Figure 21: UART Dump command with contents of sd card showing same results

4.9 RFID (functional)

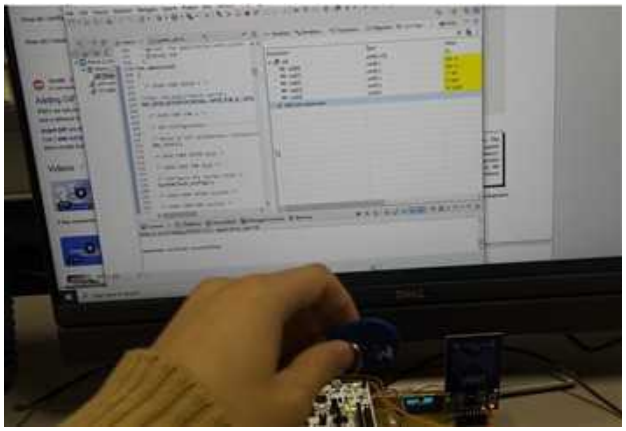


Figure 22: uid array in debug mode just before RFID

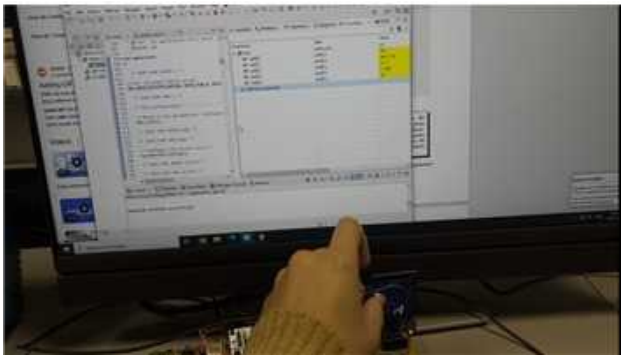


Figure 23: uid when tag present

`https://git.cs.sun.ac.za/ee_eng/design_e314/2025/26409739-e314-proj/-/blob/main/Testing/RFID_proof.MOV`

4.10 Menu Funtionality

Shown in appendix

4.11 Full system (reliable, consistent, stress testing)

5 Conclusion

5.1 Compliance non-compliance

5.2 Recommendations and Possible improvements

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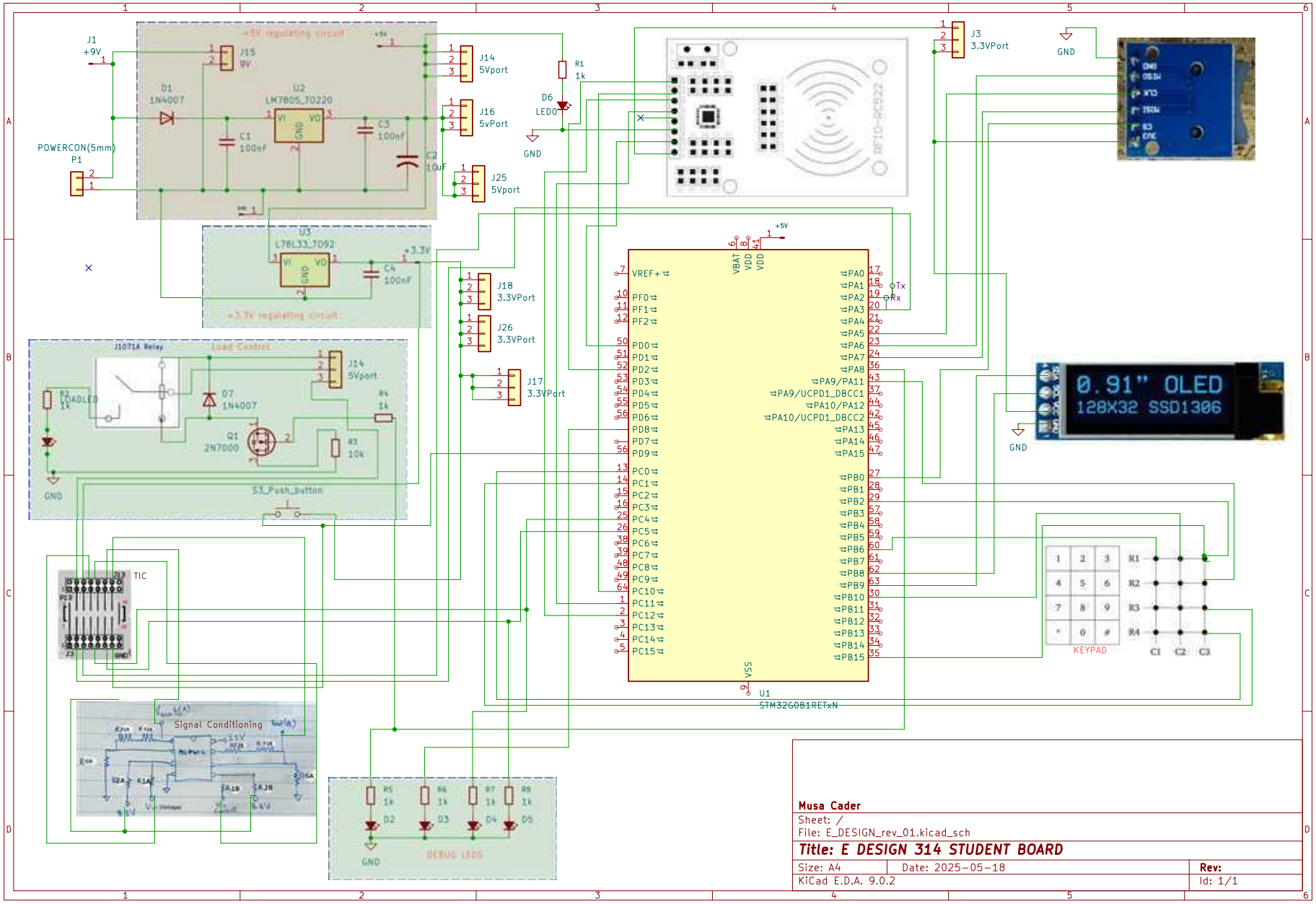
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A Complete Schematic



B STM32 pins used and configuration

C OLED MENUS

Most menus are displayed here. All menus were confirmed to work in Demo 4.



Default page



Display page 1



Display page 2



Display page 3



Enter menu 1



Load switch menu



Menu 2



Unit count menu



Count on selected



Count off selected



Menu 3



RFID authentication